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DETAILED PROPOSAL

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**Project Title:** AR-Enabled Campus Navigation and Lab Locator Application

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## **1.0 Introduction**

Universiti Putra Malaysia (UPM) is a large higher education institution comprising numerous faculty buildings, lecture halls, laboratories, administrative offices, and student facilities. Due to the complex architectural layouts and multi-floor building structures, students and visitors often encounter difficulties when attempting to locate specific destinations within campus facilities.

The Faculty of Computer Science and Information Technology (FSKTM), in particular, consists of multiple lecture rooms, laboratories, offices, and interconnected corridors that may be challenging for first-time visitors and newly enrolled students to navigate. Existing navigation methods primarily depend on physical signages, static floor plans, and verbal directions obtained from faculty staff or fellow students. These approaches frequently result in confusion, delays, and inefficiencies.

To address these challenges, this proposal introduces an AR-Enabled Campus Navigation and Lab Locator Application, a mobile-based solution that utilizes Augmented Reality (AR) technology to provide real-time indoor navigation assistance. The proposed system aims to enhance wayfinding efficiency, reduce navigation-related delays, and improve the overall campus experience for students and visitors.

## **2.0 Problem Diagnosis**

Indoor navigation remains a significant challenge in many educational institutions due to the limitations of traditional wayfinding methods. While outdoor navigation can be effectively supported through technologies such as GPS, these technologies often become ineffective once users enter building interiors.

Within faculty buildings, users frequently encounter difficulties identifying the correct floor, corridor, lecture room, laboratory, or staff office. This problem is particularly evident among first-year students and external visitors who possess limited familiarity with the campus environment.

### **2.1 Quantitative Analysis of Existing Issues**

A survey conducted among 45 students of Universiti Putra Malaysia revealed several concerns regarding current navigation practices.

The findings indicate that:

- 68.9% of respondents have experienced lateness to lectures or laboratory sessions due to difficulties locating their destinations.
- 68.9% reported moderate to high levels of difficulty in navigating faculty buildings during the initial weeks of their studies.
- 51.1% identified inadequate signage as the primary factor contributing to navigation challenges.
- 44.4% experienced stress or anxiety when attempting to locate unfamiliar locations.
- 44.4% indicated that lecture halls were among the most difficult locations to locate.
- 42.2% reported difficulties finding staff offices.

These findings demonstrate that existing navigation solutions are insufficient in supporting efficient indoor wayfinding.

## **2.2 Limitations of Existing Navigation Methods**

Several limitations have been identified in the current navigation approaches:

### **Physical Signages**

Although signboards provide basic directional guidance, they often lack detailed information and may not adequately support navigation across multiple floors.

### **Static Maps and PDF Floor Plans**

Printed maps and digital floor plans require users to interpret spatial information independently. Such resources do not provide real-time guidance and may be difficult for inexperienced users to understand.

### **Verbal Directions**

Seeking assistance from faculty staff or fellow students can be effective; however, this approach is inconsistent, time-consuming, and may interrupt staff productivity.

### **GPS-Based Applications**

Popular navigation applications such as Google Maps provide excellent outdoor navigation support but generally lack accurate indoor positioning and floor-level navigation capabilities.

### **3.0 Literature Review**

Augmented Reality (AR) refers to the integration of virtual objects and information into the real-world environment. According to Azuma (1997), AR systems possess three essential characteristics:

1. Combination of real and virtual environments.
2. Real-time interaction.
3. Three-dimensional registration of virtual objects.

AR technology has gained widespread adoption across various sectors, including healthcare, manufacturing, tourism, and education. In navigation systems, AR can provide visual overlays such as directional arrows and route indicators that guide users toward their destinations.

#### **3.1 Indoor Navigation Systems**

Indoor navigation systems are designed to support navigation within enclosed environments where traditional GPS technologies are ineffective.

Various approaches have been employed, including:

- Wi-Fi positioning systems
- Bluetooth beacon systems
- QR-code navigation systems
- Computer vision-based localization
- Augmented Reality navigation

Research has demonstrated that AR-assisted navigation enhances user comprehension, improves navigation accuracy, and reduces cognitive workload when compared to traditional map-based systems.

### **3.2 AR Navigation in Educational Institutions**

Several universities worldwide have begun exploring AR-based campus navigation systems to assist students and visitors in locating classrooms, laboratories, and administrative offices. These systems contribute to improved accessibility, increased user satisfaction, and enhanced campus digitalization initiatives.

## **4.0 Analysis of Existing Related Systems**

### **4.1 Google Maps**

Google Maps is a widely used navigation application that provides accurate outdoor navigation and real-time route guidance. However, its indoor navigation capabilities are limited, particularly in identifying specific lecture rooms, laboratories, and offices within university buildings. It also lacks detailed floor-level guidance, making it less effective for navigating complex faculty buildings.

### **4.2 Static Campus Floor Plans**

Static campus floor plans, whether printed or in PDF format, provide users with a general overview of building layouts and room locations. Although they are easy to distribute and inexpensive to maintain, they do not offer real-time guidance. Users must interpret the maps themselves, which can be challenging for first-time visitors and new students.

### **4.3 Human-Assisted Navigation**

Human-assisted navigation involves asking staff members or fellow students for directions. While this method can provide immediate assistance, it depends on the availability of others and may interrupt staff from their duties. In addition, the accuracy of directions may vary depending on the individual's knowledge of the building.

#### **4.4 Comparative Analysis**

The analysis shows that existing navigation methods have several limitations. Google Maps lacks effective indoor navigation, static floor plans do not provide real-time guidance, and human-assisted navigation is not always reliable or efficient. Therefore, an AR-based navigation system is proposed to provide real-time, accurate, and user-friendly indoor navigation within faculty buildings.

### **5.0 Proposed Treatment**

#### **5.1 Proposed Solution**

This project proposes the development of a mobile application that integrates Augmented Reality technology with an indoor positioning mechanism to provide accurate navigation assistance within university buildings.

The application will enable users to search for specific destinations, including lecture halls, laboratories, and faculty offices. Upon selecting a destination, the system will generate a route and display AR-based directional guidance through the smartphone camera interface.

#### **5.2 Operational Scenario**

A newly enrolled student intends to attend a laboratory session in Computer Laboratory 3.

Using the proposed application, the student can:

1. Open the mobile application.
2. Search for "Computer Laboratory 3".
3. Activate the AR navigation feature.
4. Follow virtual arrows displayed on the smartphone screen.
5. Arrive at the destination efficiently without requiring external assistance.



## **6.0 Objectives**

The proposed system aims to simplify indoor navigation within faculty buildings through the use of Augmented Reality technology. It will provide real-time guidance to lecture rooms, laboratories, and faculty offices, helping users locate destinations more efficiently. The system is also expected to reduce student lateness caused by navigation difficulties and minimize interruptions to faculty staff who are frequently asked for directions. Ultimately, the application seeks to enhance the overall campus experience by introducing an innovative and user-friendly navigation solution.

## **7.0 Services Provided by the System**

The proposed system will provide several services to assist users in navigating faculty buildings efficiently. The application will display real-time AR directional arrows and route indicators to guide users to their destinations. It will also include a smart room directory that allows users to search for lecture rooms, laboratories, and faculty offices. In addition, the system will support multi-floor navigation by providing guidance through staircases or elevators when users need to move between different levels. Frequently visited locations can be saved using the bookmark feature for quicker access in future visits. Furthermore, the system will automatically recalculate routes whenever users deviate from the recommended path, ensuring continuous and accurate navigation assistance.

## **8.0 Expected Business Value**

The proposed system is expected to improve the navigation experience for students, faculty staff, and visitors within campus buildings. By providing real-time indoor navigation, users can locate lecture rooms, laboratories, and offices more quickly and accurately. This can help reduce confusion, stress, and lateness among students. The system can also reduce the number of navigation-related inquiries received by faculty staff, allowing them to focus on their primary responsibilities. Furthermore, the application supports the university's Smart Campus initiative by adopting innovative AR technology and enhancing overall campus accessibility and user satisfaction. The implementation of the system is expected to significantly reduce the time required to locate destinations within faculty buildings.

## 9.0 Proposed Plan of Work

| Phase                   | Activities                                         | Duration (Week) |
|-------------------------|----------------------------------------------------|-----------------|
| Project Planing         | Background study and proposal preparation          | 8-9             |
| Detailed Proposal       | Literature review and system analysis              | 9-10            |
| Requirement Elicitation | Interviews, surveys and stakeholder analysis       | 10-12           |
| Requirement Modelling   | UML diagram and system modelling                   | 12              |
| SRS Development         | IEEE - Standard Software Requirement Specification | 13-14           |
| Final Presentation      | Report submission and presentation                 | 14              |

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